

Improved Methods for Implied Volatility Surface and Implied Distributions.

Hyuncheul Lim*

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Abstract

We present a highly mechanical method for the construction of an implied volatility surface and implied transition probability density function that satisfies the no arbitrage condition(NAC) for use in an environment with missing data environment. This paper assumes a Dupire's model which is a simple extension of the Black-Scholes model. The missing data problem for implied volatility is the first obstacle for constructing a volatility surface, the importance of the variance surface in relation to the thin plate spline(TPS) is explained.

Using a TPS approximated variance surface as a subsidiary data source, we are able to formulate a linear programming problem for the construction of the implied transition density functions, which are inherent in the market volatilities. Even though the constructed transition density function is a simple piecewise linear function, it can reconstruct the market implied volatilities extremely accurately. We faithfully showing the process leading up to the final result using helpful figures and stand-alone algorithms.

keywords: Implied volatility, Black Scholes Model, Dupire Model, Missing Data, Thin Plate Spline, Implied Distribution, Variance Surface, Linear Programming

1 Introduction

In the classical Black-Scholes model [2], unique constant volatility is assumed. However different implied volatilities are observed in the market

*Department of Mathematics, Chonnam National University, Gwangju 61186, Republic of Korea. limhc@jnu.ac.kr

depending on factors such as the maturities and exercise prices. This is reflected in practice, instead of seeing constant volatility, different volatilities are used according to each maturity and exercise price. A reinterpreted Black's Market model is represented by the SDE of the forward price $F_T(t)$ at time t with fixed maturity T :

$$\frac{dF_T(t)}{F_T(t)} = \sigma_{K,T} dW^{K,T}(t), \quad (1)$$

The subscript K, T of the volatility $\sigma_{K,T}$ indicate the value is applied in a different way according to each vanilla option with an exercise price of K and maturity T . This is the same for the superscripts K, T of the wiener process $W^{K,T}(t)$.

However, since the Black-Scholes model was built on a dynamic hedge, which in practice is important, there are attempts to account for the different implied volatilities based on a risk-neutral measure where dynamic hedges are possible. This forms a local volatility model that was developed by Dupire [5] and Derman-Kani [4]. The model (1) extends to a local volatility model (hereinafter Dupire [5]) that can evaluate complex derivatives by changing the constant interest rate and dividend to time-dependent, and constant volatility to the function of time and stock price.

$$\frac{dS(t)}{S(t)} = (r(t) - q(t))dt + \sigma_{loc}(S(t), t) dW^Q(t), \quad (2)$$

where $r(t), q(t)$ is instantaneous risk free interest rate and dividend rate, $W^Q(t)$ is the wiener process with risk neutral spot measure Q . Equation (2) represents a unified model that makes it possible to use a unique risk neutral measure for equation (1), a different model according to maturities and exercise prices.

The transition probability density function, which can be derived from Breeden's formula (5b), is obtained from the continuous call option price curve that is constructed using the different volatilities from the equation (1), is also an accompanying probability density function of the SDE (2). This is the link between the series of equations (1) and Dupire's equation (2). However, most option markets lack of liquidity to provide the necessary implied volatilities to apply equation (2). In this paper, we present a practical and highly accurate method for constructing the implied volatility surface and implied transition distributions with missing data and under violation of the NAC conditions inherent in volatilities. This paper contributes to the classical part of financial engineering in two ways. First, the missing

data problem that comes from the rareness of the market implied volatility, is applied to the estimation of the missing variance, by corresponding to minimizing the difference between the variances of each equation (1). Each market model has high correlation between them, according to the distance of its coordinates in state space, where the coordinates are maturity and exercise pair. After changing the volatility to variance, the use of a thin plate spline gives a good representation of the flatness of the variance surface and the linearity over time. This improves the fitness of the surface and the estimation of the missing data compared to using the volatility surface. Second, we obtain an intuition from the cubic smoothing spline method proposed by Matthias R. Fengler [8], and so construct the reconstruction of the transition probability density function as an optimization problem itself, then present the related linear programming problem. As a result, although the transition probability density function is a simple form of piecewise linear interpolation, it can reconstruct market volatility information very accurately. This can help to solve the numerical difficulties associated with the calibration of the local volatility, as well as aid the study of many financial engineering and finance areas that require a transition probability density function.

2 No Arbitrage Condition.

2.1 No Arbitrage Condition for Call Option

Suppose that the possible area in which the future stock price $S(T), T \geq 0$ can move by the dynamical equation (2) of the stock price S is the state space $\tilde{S} = \{(S, T) \mid S \geq 0, T \geq 0\}$ where the stock price S and the time T are coordinates.

The transition probability density function $\mathbf{p}^{\tilde{S}}(S(T), T \mid S(t), t)$ accompanying by the equation (2), represents the probability of starting from state $(S(t), t)$ and arriving at state $(S(T), T)$. Let $\mathbf{P}^{\tilde{S}}(S(T), T)$ be the case where the starting point $(S(t), t)$ is the origin $(S(0), 0)$.

When the volatility on the generated surface satisfies the NAC condition, it means that the price of call options with the same underlying evaluated by this volatility satisfies the no arbitrage condition. Let's list the NAC conditions by converting the volatility to the call option price.

If we assume a fixed volatility σ is applied, the transition probability $\mathbf{p}^{\tilde{S}}$

have the following analytic formula.

$$\mathbf{p}^{\tilde{S}}(S(T), T | S(t), t) = \frac{1}{S(T)\sigma\sqrt{2\pi\tau}} \exp\left(-\frac{(\ln S(T) - \ln S(t) - \bar{\mu})^2}{2\sigma^2\tau}\right), \quad (3)$$

where $\bar{\mu} = (\bar{r} - \bar{q} - \frac{1}{2}\sigma^2)\tau$, $\tau = T - t$.

Assume the Dupire's model (2) and its accompanying transition density function (3), we can then represent the call option price $C(K, T; t)$ at time $t < T$ by $\mathbf{p}^{\tilde{S}}$ (3).

$$C(K, T, t) = df \int_0^\infty \max(S(T) - K, 0) \mathbf{p}^{\tilde{S}}(S(T), T | S(t), t) dS(T) \quad (4)$$

Theorem 2.1 (NAC for Call Option). *The conclusions obtained from the call option price formula (4) obtained by assuming the existence of the transition probability density function $\mathbf{p}^{\tilde{S}}$ (3) under the Dupire model (2):*

$$\circ \quad -df \leq \frac{\partial C}{\partial K} < 0, \quad (5a)$$

$$\circ \quad \frac{\partial^2 C}{\partial K^2} = df \cdot \mathbf{p}(S(T), T | S(t), t) > 0, \quad (5b)$$

$$\circ \quad dq C(K_1, T_1; t) \leq C(K_2, T_2; t), \quad T_1 < T_2, \quad K_2 = \frac{dq}{df} K_1, \quad (5c)$$

$$\circ \quad \max\left(dq S(t) - df K, 0\right) \leq C(K, T, t) \leq dq S(t), \quad (5d)$$

where $df = e^{-\int_t^T r(s)ds}$, $dq = e^{-\int_t^T q(s)ds}$.

Proof. Differentiate the expression (4) once for K to obtain

$$\frac{\partial C}{\partial K} = df \cdot (\text{cdf}(K) - 1), \quad (6)$$

where $0 \leq \text{cdf}(K) = \int_0^\infty \mathbf{P}^{\tilde{S}}(S(T), T) dS(T) < 1$. Thus formula (5a) $-df \leq \frac{\partial C}{\partial K} < 0$ is established. Breeden-Litzenberger(1978) [3] obtained equation (5b) by differentiation of (6) $\frac{\partial^2 C}{\partial K^2} = df \cdot \mathbf{p}^{\tilde{S}}(S(T), T | S(t), t) > 0$.

Let $\tilde{S}(t) = S(t)e^{-\int_0^t r(s)-q(s)ds}$, $t \geq 0$ then under the risk-neutral Q measure induced by (2), $\tilde{S}(t)$ is martingale i.e under filtration $\mathcal{F}_t$ for $t > 0$,

$\mathbb{E}[\tilde{S}(T) | \mathcal{F}_t] = \tilde{S}(t)$ for $T > t \geq 0$. More precisely, condition (5c) holds directly from the following inequality which uses the martingale result,

$$\begin{aligned} \mathbb{E} \left[\max(\tilde{S}(T_2) - K, 0) | \mathcal{F}_{T_0} \right] &= \mathbb{E} \left[\mathbb{E}[\max(\tilde{S}(T_2) - K, 0) | \mathcal{F}_{T_1}] | \mathcal{F}_{T_0} \right] \\ &\geq \mathbb{E} \left[\max \left(\mathbb{E}[\tilde{S}(T_2) | \mathcal{F}_{T_1}] - K, 0 \right) | \mathcal{F}_{T_0} \right] \\ &\geq \mathbb{E} \left[\max \left(\tilde{S}(T_1) - K, 0 \right) | \mathcal{F}_{T_0} \right] \end{aligned} \quad (7)$$

where the first equality holds for iterated expectation, the second inequality for convexity of the payoff function and the third inequality holds for a martingale result.

The left hand inequality of condition (5d) holds using (7) with $T_2 = T$ and $T_1 = 0$. The right hand side of inequality holds for (5a), $C(K, T; t) < C(0, T; t)$, $K > 0$ and $\max(S(T), 0) = S(T)$. $\square$

2.2 Transformation of state space to forward moneyness

In order to eliminate the effect of interest rates($r(t)$) and dividend rates($q(t)$) and to standardize the spot and exercise price, we use the forward moneyness as the underlying asset axis of the state space. According to [5], [7], [8], the forward moneyness is given by

$$x = \frac{K}{F_T}, \text{ where } F_T = S_0 e^{\int_0^T (r(s) - q(s)) ds} \quad (8)$$

Now the original state space $\tilde{\mathcal{S}} = \{(K, T) | K > 0, T > 0\}$ transforms to $\mathcal{S} = \{(x, t) | x > 0, t > 0\}$ with the relations $x = \frac{K}{F_T}$ and $t = T$.

The Black-Scholes call option price $Call(K, T)$ with strike K and maturity T in $\tilde{\mathcal{S}}$ corresponds to $call(x, t)$ in $\mathcal{S}$ with the following relation.

$$\begin{aligned} Call(K, T) &= D_T [F_T N(d_1) - K N(d_2)] \\ &= D_T F_T [N(d_1) - x N(d_2)] \\ &= D_T F_T call(x, t) \end{aligned} \quad (9)$$

where $D_T = e^{-\int_0^T r(s) ds}$, $x = \frac{K}{F_T}$, $t = T$, $d_1 = \frac{\ln(\frac{1}{x})}{\sigma\sqrt{t}} + \frac{1}{2}\sigma\sqrt{t}$, $d_2 = d_1 - \sigma\sqrt{t}$. From this relation, in variance surface coordinates (x, t, u) the standard call option price $call(x, t, u)$ is

$$call(x, t, u) = N(d_1) - x N(d_2) \quad (10)$$

where $x = \frac{K}{F_t}$, $d_1 = \frac{\ln(\frac{1}{x})}{\sqrt{u}} + \frac{1}{2}\sqrt{u}$, $d_2 = d_1 - \sqrt{u}$, $u = \sigma^2 t$

Theorem 2.2 (NAC for Calendar Spread). *For the variance surface in $\tilde{\mathcal{S}}$, one of the conditions of the NAC is the calendar arbitrage is directly seen in the surface as follows.*

$$u(x, t_1) < u(x, t_2), \quad \text{for all } x, t_1, t_2 \text{ s.t. } t_1 < t_2. \quad (11)$$

Proof. This relation is easily checked so

$$0 < \frac{\partial \text{call}(x, t, u)}{\partial t} = \frac{\partial \text{call}(x, t, u)}{\partial u} \frac{\partial u}{\partial t} = \frac{1}{2} \frac{n(d_2)}{\sqrt{u}} \frac{\partial u}{\partial t} \quad (12)$$

where $n(z) = \frac{dN(z)}{dz}$ and $\frac{n(d_2)}{\sqrt{u}} > 0$, $\frac{\partial u}{\partial t}$ must be greater than 0. $\square$

Now let's reformulate the Theorem 2.1 from transformed state space $\mathcal{S}$. However, unlike in the previous theorem, this is limited to starting from the origin $(1, 0)$.

Theorem 2.3 (NAC for Call Option under $\mathcal{S}$).

$$\circ \quad -1 \leq \frac{\partial c}{\partial x} < 0, \quad (13a)$$

$$\circ \quad \frac{\partial^2 c}{\partial x^2} = \mathbf{p}(x, t | 1, 0) > 0, \quad (13b)$$

$$\circ \quad c(x, t_1) \leq c(x, t_2), \quad t_1 < t_2, \quad (13c)$$

$$\circ \quad \max(1 - x, 0) \leq c(x, t) \leq 1, \quad (13d)$$

In the next section, we demonstrate that TPS approximation for the variance surface is a very efficient tool for solving the missing data problem in the market volatility. Also we show the two examples and a simple but concrete algorithm for solving the problem.

3 Variance Surface: The Pre-Smoother.

3.1 Radial Basis Function and Thin Plate Spline.

In the options market, it is important to estimate the missing volatilities for pricing and hedging the derivatives. In some cases, it is more efficient to approximate a surface consisting of the variance or call option prices

instead of volatility itself. This article use a variance surface for filling the missing data and a call option price surface for satisfying the NAC. The most popular method of approximating a surface like this is with cubic splines, but this needs gridded data, i.e. data that is located on a regularly spaced mesh. However in the society of statistical data fitting, the use of radial basis function(RBF) have been studied and applied.

We introduce a brief theoretical description of the radial basis function including TPS approximation in appendix 5.6. This approach is very effective for fitting the scattered data because of the distance-dependent nature of the basis function. Among these kinds of functions, thin plate splines (TPS) have been studied in depth by Wahba et al [12]. In particular, in the area of financial derivatives, the TPS has been applied to the implied volatility surface generation by Orosi [22]. Part of the reason for this is the basis function of the TPS is a fundamental solution of a biharmonic equation, this is the equation for minimizing the bending energy of a surface. (41)

A variance surface aims to fillin the missing data using the linearity property of the variance along its time axis as well as the minimal bending property of TPS. This variance surface provides additional volatilities that are needed for generating the call option price curves and implied distributions which are our main target.

3.2 Variance Surface using TPS.

Far point regions from the at the money lead month options, which is the point $(1, t_1)$ in state space $\mathcal{S}$, which are deep OTM put, deep OTM call and of longer maturity are areas where the traded data is very rare, as shown in Figure 1. In the general RBF approximation which uses the input data locations as the center of the basis functions, the estimation of the far point area has a strong tendency to bend towards the existing data cluster. Therefore, when approximating a surface which is supported on a square region 1, some of the boundary points of the supporting square are necessary to frame the surface. TPS gives a surface with minimal bending and, unlike other RBFs, it does not need the shape parameter ϵ . In particular, when the existing variances are clustered and forms a shape like a flat plane, the approximated far point value is located near the extended plane of the clustered data compared results from other RBFs (eg: Gaussian).

3.3 Boundary Data Estimation

If the data in the far point region in figure 1 is very scarce, artificial data should be added to help frame the variance surface. Three methods are available to do this.

- Smoothness Parameter p .

Since the TPS approximation has no shape parameter, the smoothness parameter p plays an important role. If p is close to 1, the local effect of individual observations is strong. This difference has a more distorted effect in the region of no data. In the case of our data settings, the average effect is highlighted at $0.9 \leq p \leq 0.95$, which gives a good representation of the overall surface appearance, although the accuracy of the observation is relatively poor than exact fit. In this case, the generated variance surface is smoothed so that it does not need artificial boundary data. We capture this boundary data as added data for more accurate fitting for changing $p \rightarrow 1$.

- Heuristic Decision Rule

Many traders have their decision rule for far point volatility values, either by comparing with the average level of the traded option prices or by referring to the current trends. In view of this, it is also possible to follow a heuristic (legacy) internal decision rule.

- Regression Method

This method, previously called the practitioners volatility surface model, was used in other papers to approximate the implied volatility surface. However, linear regression severely lacks of fitting due to the smile and skew of the implied volatility surface. Therefore, we target the variance surface for regression. In addition, separately regress the each portion of the area instead of the whole, provide more reasonable results.

3.4 Algorithm for Drop Out Calendar Arbitrage.

Calendar Arbitrage often occurs when there are very short maturity and time mismatches in data capturing. If we allow these, the surface have distorted shape and the sequential calibration of call option price curves is impossible.¹

In practice, the construction process for the call option price curve could corrects the errors for three kinds of NAC, in most cases, except for the calendar arbitrage. Thus the abnormal data, which causes a calendar arbitrage, must be extracted by modification or removal.

We have tried many methods such as leave one out cross validation [22],

¹The previous price curve, which is a lower bound for current price curve, prevents normal solution curve.

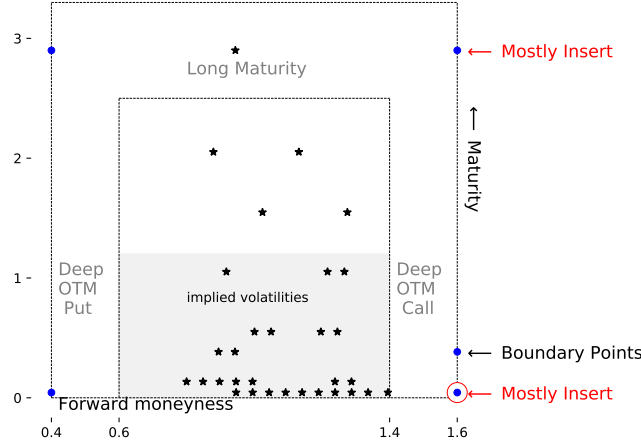


Figure 1: $\star$'s are scattered volatilities in the market and $\bullet$'s are artificially inserted boundary data. The observed volatilities are clustered in the gray colored area. The bottom right point is especially important.

generalized cross validation [22], which are well known method for RBF approximation [16], and applying the quadratic programming method using the basis of TPS,² but no other methods including these, are not fully reliable except direct simple search method. Let's describe this simple search strategy for extracting the abnormal volatility that causes a calendar arbitrage.

We set $p = 1$ to reflect market volatility as much as possible, excluding unusual data we are looking for. Let the set $\mathbf{X} = \{X_1, \dots, X_{N_x}\}$ is unique x_i 's in Ω_N and $\mathbf{T} = \{T_1, \dots, T_{N_t}\}$ uniquely sorted $t_i, i = 1, \dots, N$. For each X_i with corresponding T_j , when $1 < j < N$, calculate two different variances $f_{TPS}(X_i, T_{j-1} | \Omega_N, p)$ and $f_{TPS}(X_i, T_{j+1} | \Omega_N, p)$.

If both of the conditions $f_{TPS}(X_i, T_{j-1} | \Omega_N, p) < f_{TPS}(X_i, T_j | \Omega_N, p)$ and $f_{TPS}(X_i, T_j | \Omega_N, p) > f_{TPS}(X_i, T_{j+1} | \Omega_N, p)$ holds, then (X_i, T_j) which is k -th data $\mathbf{x}_k \in \Omega_N$ is the strong candidate for calendar arbitrage.

Also **when** $j = 1$ and $f_{TPS}(X_i, T_1 | \Omega_N, p) > f_{TPS}(X_i, T_2 | \Omega_N, p)$ holds, then $(X_i, T_1) = \mathbf{x}_k$, and lastly **for** $j = N$, and $f_{TPS}(X_i, T_N | \Omega_N, p) < f_{TPS}(X_i, T_{N-1} | \Omega_N, p)$ holds, then $(X_i, T_N) = \mathbf{x}_k$ is.

² $\min \sum_{i=1}^N (f_{TPS}(x_i, t_i) - y_i)^2$ with constraints $f_{TPS}(x_i, t_i) > 0, \frac{\partial f_{TPS}(x_i, t_i)}{\partial t} > 0$ forms a QP problem, using the relation (43), but conditions are locally satisfied only at the points $(x_i, t_i), i = 1, \dots, N$. It does not work globally.

Remove this from Ω_N and retry the TPS with $\Omega = \Omega_N - \mathbf{x}_k$. A detailed algorithm is described in the FCA function **FCA** of the algorithm part 1, where Ω denote the dataset after processing and $f_{TPS}(\mathbf{x} | \Omega, p)$ be our final pre-smoothing variance surface. A few steps of this procedure gives us the best result among all other methods. In our example in table 1, and figure 2, 4, shows this process.

Algorithm 1: Function: FCA

Data: $\Omega_N = \{(x_i, u_i) \mid i = 1, \dots, N, \mathbf{x}_i = (x_i, t_i), u_i = \sigma_i^2 t_i\}, p \leftarrow 1$

Result: $\mathcal{D}$ is the set of elements which violate the monotone increasing as a function condition of t with given fixed moneyiness $f_{TPS}(\cdot, t)$.

Function FCA(Ω_N):

```
 $\mathcal{D} = []$ 
 $\mathbf{X} = [X_1, \dots, X_{N_x}] \leftarrow \text{sort}(\text{unique}([x_i, i = 1, \dots, N]))$ 
 $\mathbf{T} = [T_1, \dots, T_{N_t}] \leftarrow \text{sort}(\text{unique}([t_i, i = 1, \dots, N]))$ 
for  $X_i, i = 1, \dots, N_x$  do
    find  $T_j \in \mathbf{T}$  where  $\mathbf{x}_k = (X_i, T_j) \in \Omega_N$ 
     $\tilde{u}_j \leftarrow f_{TPS}(X_i, T_j \mid \Omega_N, p)$ 
    if  $1 < j < N_t$  then
         $\tilde{u}_{j-1} \leftarrow f_{TPS}(X_i, T_{j-1} \mid \Omega_N, p)$ 
         $\tilde{u}_{j+1} \leftarrow f_{TPS}(X_i, T_{j+1} \mid \Omega_N, p)$ 
        if  $\tilde{u}_{j-1} < \tilde{u}_j$  and  $\tilde{u}_j > \tilde{u}_{j+1}$  then
             $\mathcal{D} = \mathcal{D} \cup (\mathbf{x}_k, u_k)$ 
    else if  $j = 1$  then
         $\tilde{u}_2 \leftarrow f_{TPS}(X_i, T_2 \mid \Omega_N, p)$ 
        if  $\tilde{u}_1 > \tilde{u}_2$  then
             $\mathcal{D} = \mathcal{D} \cup (\mathbf{x}_k, u_k)$ 
    else if  $j = N_t$  then
         $\tilde{u}_{N_t-1} \leftarrow f_{TPS}(X_i, T_{N_t-1} \mid \Omega_N, p)$ 
        if  $\tilde{u}_{N_t-1} > \tilde{u}_{N_t}$  then
             $\mathcal{D} = \mathcal{D} \cup (\mathbf{x}_k, u_k)$ 
return  $\mathcal{D}$ 
```

$\Omega = \Omega_N - \mathcal{D}$

reset: $N \leftarrow \text{len}(\Omega_N) - \text{len}(\mathcal{D})$

redefine: $\Omega = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$

Function FTPS($\mathbf{x}, \Omega, p$):

```
 $\mathbf{c} = (c_1, \dots, c_{N+3})$  from  $\Omega, p$  and algorithm 4
 $f = c_{N+1}x + c_{N+2}t + c_{N+3}$ 
for  $\mathbf{x}_i, i = 1, \dots, N$  do
     $\mathbf{r}_i = \|\mathbf{x}_i - \mathbf{x}\|$ 
     $f = f + 2c_i \mathbf{r}_i^2 \ln \mathbf{r}_i$ 
return  $f$ 
```

Table 1: Market implied volatilities, rows are moneyness, [red] values are artificially inserted abnormal volatilities. Blue colored volatilities are boundary data for the frame of surface. HSCEI VOLATILITY 09/11/2015, Spot: 10245.53

Maturity(days)	16	49	140	201	384	565	749	1114	1876
ZeroRate	0.0038	0.0038	0.0047	0.0056	0.0069	0.0085	0.0102	0.0129	0.0168
DividendRate	-0.00968	-0.00286	-0.00645	0.0146	0.0271	0.0268	0.0270	0.0292	0.0284
0.50	0.5	-	-	-	-	-	-	-	0.5
0.65	-	-	-	-	-	-	-	-	0.29
0.80	0.4563	0.364	-	-	-	-	-	-	-
0.85	-	0.3333	-	-	-	-	0.2577	-	-
0.90	0.3555	0.2997	0.2787	-	0.2551	-	-	0.2515	-
0.95	0.3005	0.2722	0.2625	-	-	-	-	-	-
1.00	0.2772	0.2607	-	0.2435	-	0.2399	-	-	-
1.05	0.288	-	-	0.2374	-	-	-	-	-
1.10	0.3049	-	-	-	-	-	0.2361	-	-
1.15	0.3184	-	-	-	-	-	-	[0.4]	-
1.20	0.3297	-	-	0.2422	0.2349	-	-	-	-
1.25	0.3405	0.2924	-	0.2478	0.2378	0.235	-	-	-
1.30	0.351	0.3013	-	-	-	-	-	-	-
1.35	0.3611	-	[0.5]	-	-	-	-	-	-
1.40	0.3708	-	-	-	-	-	-	-	0.24
1.45	-	-	-	-	-	-	0.2408	-	-
1.50	-	-	-	0.2786	-	-	-	-	0.24

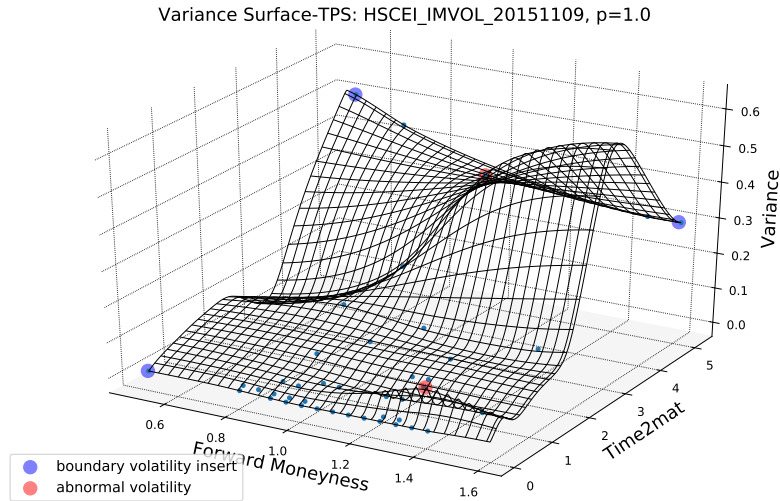


Figure 2: Variance Surface-TPS($p = 1$) with abnormal data, surface is wiggled from these two volatilities. HSCEI, 09/11/2015

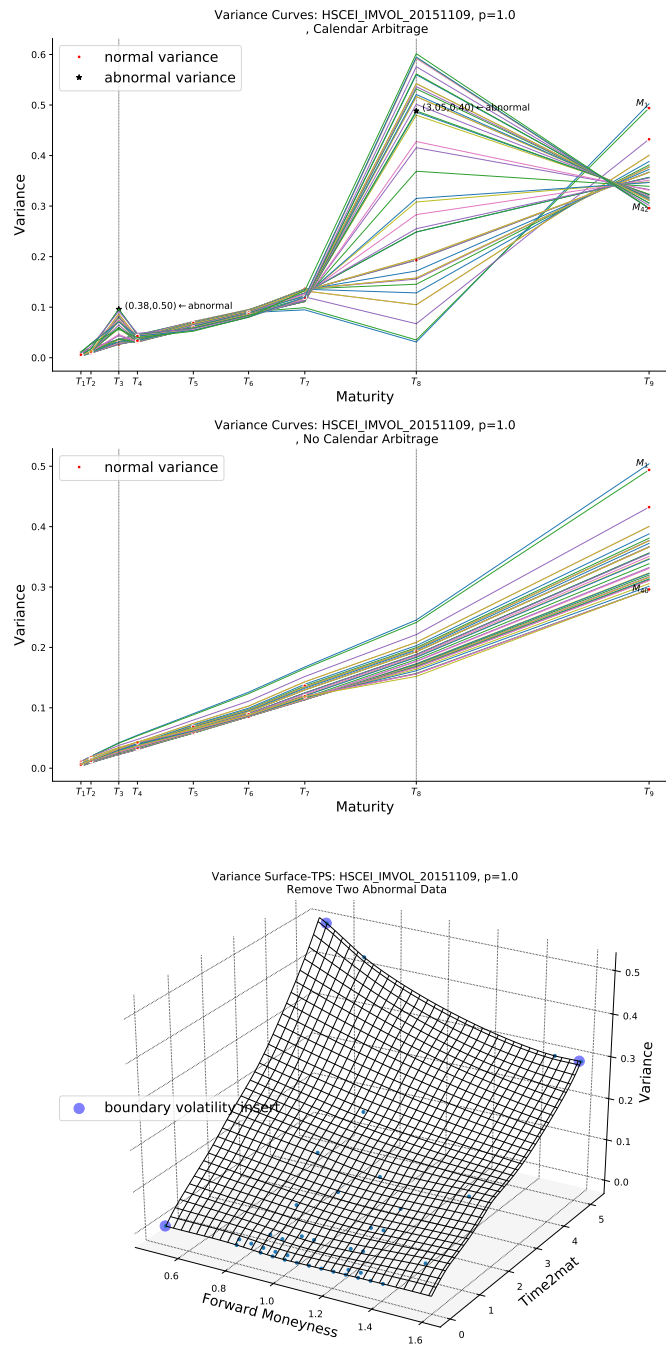


Figure 3: Direct Drop Out: TPS with $p = 1$ (figure 2), direct search every variance curve for each moneyness(total 42), which conflicts with monotone increasingness (1st figure), we removes these with algorithm 1 and retry TPS($p = 1$) (2nd and 3rd figure). HSCEI, Date: 09/11/2015

Table 2: S&P500 VOLATILITY 07/11/2019, Spot: 3085.18

Maturity(days)	43	253	407	561	771	1107	1499	1870
ZeroRate	0.01708	0.01795	0.01727	0.01685	0.01633	0.01604	0.01592	0.01598
DividendRate	0.01510	0.01648	0.01668	0.01645	0.01626	0.01627	0.01633	0.01649
0.50	-	-	-	0.30484	-	0.27	-	0.24
0.55	0.41464	0.33913	0.30897	-	0.27159	-	-	0.23529
0.70	-	0.2768	-	-	-	-	-	-
0.75	0.31032	-	-	-	-	-	-	-
0.80	0.27512	-	-	-	-	-	-	-
0.85	0.23864	0.2159	-	-	0.20524	-	-	-
0.90	-	-	0.19483	-	-	-	-	-
0.95	0.1601	0.17415	-	0.1828	-	-	-	0.19261
1.00	0.11289	-	-	-	-	-	-	-
1.05	0.09399	0.13216	-	-	-	0.17563	0.18099	0.18424
1.10	0.10868	-	0.13537	-	-	-	-	-
1.15	0.11499	0.10484	-	-	-	0.16327	-	-
1.20	0.12247	0.10142	-	-	-	-	-	-
1.25	0.12996	-	-	-	-	-	-	-
1.30	-	-	-	-	0.13027	-	-	-
1.45	0.13	-	-	-	-	-	-	-
1.50	0.13	0.1	0.105	0.112	0.11482	-	-	0.154

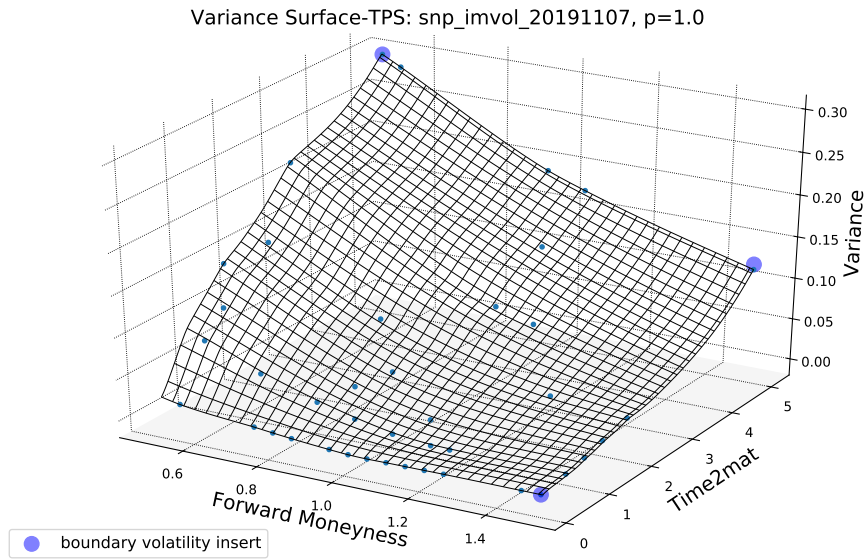


Figure 4: Variance Surface(TPS, $p = 1$): 3 boundary points(blue color) are inserted. S&P, 07/11/2019

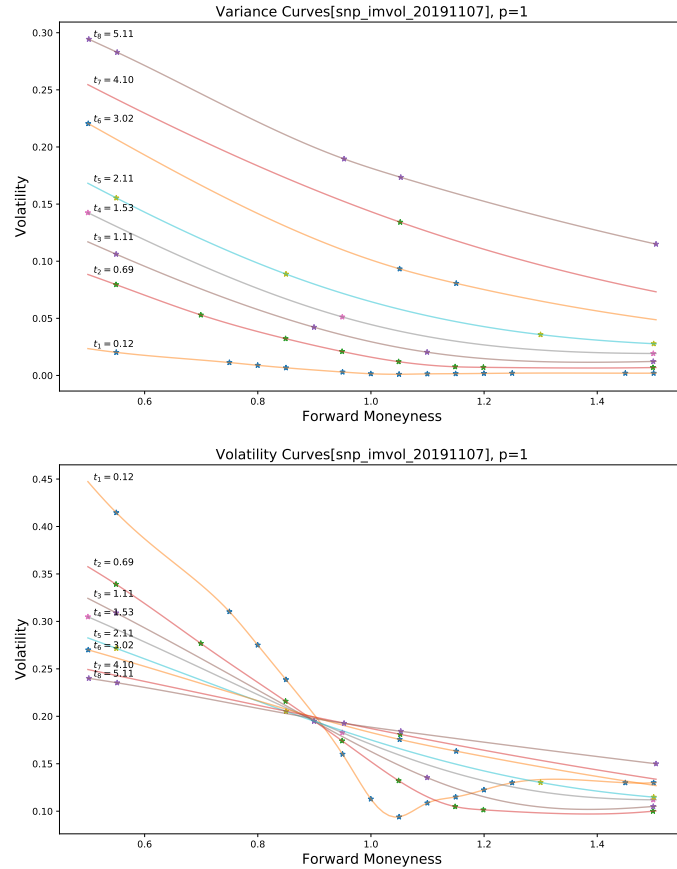


Figure 5: TPS($p = 1$) approximated variance(upper), volatility(lower) curves with respect to each maturities. These curves are exactly fitted(solid line) that give no differences compared to market values($\star$). S&P500, Date: 07/11/2019

4 Call Option Price Curves.

4.1 Least Squares Cubic Spline approximation.

Let $f(x)$ be a cubic spline polynomial with knots $x_1, \dots, x_N$,

$$f(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3, \quad (14)$$

where $x \in [x_i, x_{i+1}]$, $i = 1, \dots, N$. When at the end points x_1, x_N , forcing the condition of $c_1 = c_N = 0$ is called natural cubic spline. From Carl

De Boor [17] ch. XIV, this condition can be written as a relation between $\mathbf{a} = (a_1, \dots, a_N)^\top$ and $\mathbf{c} = (c_1, \dots, c_N)^\top$, in matrix form,

$$\mathbf{R}\mathbf{c} = 3\mathbf{Q}^\top \mathbf{a}, \quad (15)$$

where $c_1 = c_N = 0$, $\mathbf{R}$ be a symmetric tridiagonal matrix of dim $N \times N$ with tridiagonal elements of $i = 2, \dots, N-1$ rows are $\Delta x_{i-1}, 2(\Delta x_{i-1} + \Delta x_i), \Delta x_i$, which is strictly diagonally dominant³, and $\mathbf{Q}^\top$ is the $N \times N$ tridiagonal matrix with tridiagonal elements of $i = 2, \dots, N-1$ rows are $\frac{1}{\Delta x_{i-1}}, -\frac{1}{\Delta x_{i-1}} - \frac{1}{\Delta x_i}, \frac{1}{\Delta x_i}$ and all other elements of $\mathbf{R}, \mathbf{Q}$ are 0.

The smoothing spline method is a spline approximation that gives a penalty term and allows for some error from the observations. Most general form of the penalty is by minimizing the square of second derivatives over all of the range, and spline function by natural cubic spline.

$$\mathbf{J}(f) = p \sum_{i=1}^N [w_i(y_i - f(x_i))]^2 + (1-p) \int_{x_1}^{x_N} (f''(x))^2 dx, \quad (16)$$

where $y_i = g(x_i) + \epsilon_i$ for some smooth function g at data sites $x_1, \dots, x_N$, and let $\frac{1}{w_i}$ is the standard deviation in y_i for estimation errors ϵ_i , $\lambda = \frac{1-p}{p}$ is called roughness of penalty. Because $f''(x)$ is a linear function, the integral can be calculated using Simpson's rule,

$$\int_{x_1}^{x_N} (f''(x))^2 dx = \frac{4}{3} \sum_{i=1}^{N-1} \Delta x_i (c_i^2 + c_i c_{i+1} + c_{i+1}^2) = \frac{2}{3} \mathbf{c}^\top \mathbf{R} \mathbf{c}, \quad (17)$$

Now we can construct the quadratic programming problem for finding $\mathbf{a}$. From equation (15) we can express $\mathbf{c}$ using $\mathbf{a}$,

$$\mathbf{c} = 3 \left(\mathbf{R} \backslash \mathbf{Q}^\top \right) \mathbf{a} = 3\mathbf{S}\mathbf{a}, \quad (18)$$

where $\mathbf{S} = \mathbf{R} \backslash \mathbf{Q}^\top$ and the backslash operator " $\backslash$ " implies solving the matrix equation $\mathbf{R}\mathbf{S} = \mathbf{Q}^\top$ without using the inverse and ignoring the first and last rows of the $N \times N$ matrix $\mathbf{S}$ which are all 0. The integral part is $\frac{2}{3} \mathbf{c}^\top \mathbf{R} \mathbf{c} = 6\mathbf{S}^\top \mathbf{R} \mathbf{S} = 6\mathbf{M}$ where $\mathbf{M} = \mathbf{S}^\top \mathbf{R} \mathbf{S}$. Then the equation (16) can be transformed to matrix form:

$$\mathbf{J}(f)/p = \mathbf{a}^\top (\mathbf{W} + 6\lambda \mathbf{M}) \mathbf{a} - 2\mathbf{y}^\top \mathbf{W} \mathbf{a} + \mathbf{y}^\top \mathbf{W} \mathbf{y}, \quad (19)$$

³Clear from the expression, in this case $\mathbf{R}$ is strictly positive definite and the matrix equation (15) has a unique solution $\mathbf{c}$.

where $\mathbf{W}$ is the diagonal matrix with elements $w_1, \dots, w_N$. As the last term is constant, the first two terms are the objective function of the minimization problem. Since at the knots $x_1, \dots, x_{N-1}$, the value, first and second derivatives of $f(x)$ must be the same by left and right segments,

$$\begin{aligned} b_i &= \frac{1}{\Delta x_i}(a_{i+1} - a_i) - \frac{\Delta x_i}{3}(2c_i + c_{i+1}), \\ d_i &= \frac{1}{3\Delta x_i}(c_{i+1} - c_i), \quad i = 1, \dots, N-1, \end{aligned} \quad (20)$$

At x_N , $b_N = f'(x_N) = \lim_{x \rightarrow x_N-} f'(x_N) = \frac{1}{\Delta x_{N-1}}(a_N - a_{N-1}) + \frac{\Delta x_{N-1}}{3}(2c_N + c_{N-1})$ and additional assumption of $d_N = 0$. In matrix form,

$$\mathbf{c} = 3\mathbf{S}\mathbf{a}, \quad \mathbf{b} = \mathbf{B}\mathbf{a}, \quad \mathbf{d} = \mathbf{D}\mathbf{a}, \quad (21)$$

where $\mathbf{B} = \mathbf{B}_1 - \mathbf{B}_2\mathbf{S}$ and $\mathbf{D} = \mathbf{B}_1\mathbf{S}$. The $\mathbf{B}_1$ is the $N \times N$ matrix with elements $-\frac{1}{\Delta x_i}, \frac{1}{\Delta x_i}, i = 1, \dots, N-1$ in its diagonal part and $\mathbf{B}_2$ is the $N \times N$ matrix with elements $2\Delta x_i, \Delta x_i, i = 1, \dots, N-1$ in its diagonal part.

4.2 Fengler's Quadratic Optimization.

Matthias R. Fengler [8] constructs a call option price curves by a least squares approximation with smoothing cubic spline and NAC constraints. We can rewrite this problem under the state space $\mathcal{S}$ as,

$$\begin{aligned} \min_{f_j} & \left(\sum_{i=1}^{N_j} [w_{i,j}(y_{i,j} - f_j(x_{i,j}))]^2 + \lambda \int_{x_{1,j}}^{x_{N_j,j}} (f_j''(x))^2 dx \right), \text{ for } j = M, \dots, 1, \\ \text{s.t. } & f_j(x_{N_j,j}) \geq 0, \\ & \frac{\partial f_j}{\partial x} \Big|_{x_i} \leq 0, \\ & \frac{\partial^2 f_j}{\partial x^2} \Big|_{x_{i,j}} \geq 0, \quad i = 1, \dots, N_j, \\ & f_j(x_{i,j}) < f_{j+1}(x_{i,j}), \end{aligned} \quad (22)$$

where $x_{i,j} = (x_i, t_j)$ is the location of i -th moneyness x_i at maturity t_j , $y_{i,j}$ which is the actual or approximated⁴ call option price at $x_{i,j}$, $f_j(x) = a_{i,j} +$

⁴Using the pre-smoothing volatility surface.

$b_{i,j}(x-x_{i,j})+c_{i,j}(x-x_{i,j})^2+d_{i,j}(x-x_{i,j})^3$, $x \in (x_{i,j}, x_{i+1,j}]$, $i = 1, \dots, N_j$ for each t_j , $\lambda = \frac{1-p}{p}$ is the roughness of the penalty, $x_{N_j,j}$ is large enough for all $\epsilon > 0$ such that $\int_{x_{N_j,j}}^{\infty} f_j(x)dx < \epsilon$, the same for x_1 , which is small enough. This sequential optimization problem (22) proceeds from last maturity t_M to t_1 .

5 Implied Transition Density.

5.1 Constraints for Transition Density.

The Fengler's method works well when there are enough volatilities which do not violate the calendar arbitrage. The great advantage of this approach is that it can deal with the values, first and second order differential conditions in the entire range of moneyness by using low-order polynomials that are easy to handle. We describe a technical method for extracting the transition probability density function inherent in market volatility information based on the cubic splines applied in Fengler's method.

In order to extract the transition probability density function, in addition to the four NAC conditions for the call option price curve, the transition probability density function itself must additionally satisfy two conditions. Additionally, artificial but reasonable conditions are needed to reduce the peakness that the function has in the knots.

5.1.1 Probability Density Function.

Since $\frac{\partial^2 C}{\partial x^2} = \mathbf{P}(x, T)$ from Breeden and Litzenberger(1978) [3]'s formula, $\int_0^{\infty} \mathbf{P}(x, T)dx = 1$ holds. We additionally add the following constraint to (22),

$$\int_{x_1}^{x_N} f''(x)dx = 1, \quad (23)$$

where $f(x)$ is call option price curve at maturity T , $x_1 = 0$ and $\int_{x_N}^{\infty} f''(x)dx < \epsilon$ for some $\epsilon > 0$. Considering that the second order differentiation of the cubic spline function f is a piecewise linear function, the equation(23) is easily added as a constraint for quadratic optimization equation (22) by the trapezoidal quadrature.

$$\int_{x_1}^{x_N} f''(x)dx = 2\mathbf{F}^T \mathbf{c} = 6\mathbf{F}^T \mathbf{S} \mathbf{a} = 1, \quad (24)$$

where $\mathbf{F} = \frac{1}{2} (\Delta x_1, (\Delta x_1 + \Delta x_2), \dots, (\Delta x_{N-2} + \Delta x_{N-1}), \Delta x_{N-1})^T$ and $f''(x_i) = 2c_i$, $i = 1, \dots, N$.

The equality constraint (23) you want to add, greatly reduces the feasible region of the solution space. More specifically, when the equality constraint of $\int_{x_1}^{x_N} f''(x)dx = 1$ and the penalty term $\lambda \int_{x_1}^{x_N} (f''(x))^2 dx$ exist at the same time, it is difficult to find a solution for the quadratic programming (22), within an acceptable error range in most of the cases. Therefore, the penalty term of the objective function (22) must be removed or $\lambda = 0$, the good looking but sad results of this are shown in figure 15 and 16.

Even though we apply the Breeden's formula (5b), its effect depends on the subdivision of finite knots $\mathbf{x} = (x_1, \dots, x_N)$, of course it is not practical to use a dense grid of $\mathbf{x}$. Therefore, it is necessary to add more constraints to f'' which effect the entire range of the forward moneyness $\mathbf{x}$.

5.1.2 Risk Neutral Assumption.

The risk neutral assumption of the model (2), which holds in forward moneyness space $\mathcal{S}$, is simply,

$$\mathbb{E}[X] = \int_{x_1}^{x_N} x f''(x) dx = 1, \quad (25)$$

where $X = \frac{S(T)}{F_T}$, F_T in equation (8).

Since $\int_{x_i}^{x_{i+1}} x f''(x) dx = \int_{x_i}^{x_{i+1}} x(2c_i + 6d_i(x - x_i)) dx = (x_{i+1}^2 - x_i^2)c_i + (2x_{i+1}^3 + x_i^3 - 3x_i x_{i+1}^2)d_i$ and $\mathbf{c} = 3\mathbf{S}\mathbf{a}$, $\mathbf{d} = \mathbf{D}\mathbf{a}$, equation (25) becomes,

$$\begin{aligned} \int_{x_1}^{x_N} x f''(x) dx &= (x_2^2 - x_1^2, x_3^2 - x_2^2, \dots, x_N^2 - x_{N-1}^2, 0) \cdot \mathbf{c} \\ &+ (2x_2^3 + x_1^3 - 3x_1 x_2^2, 2x_3^3 + x_2^3 - 3x_2 x_3^2, \dots, 2x_N^3 + x_{N-1}^3 - 3x_{N-1} x_N^2, 0) \cdot \mathbf{d} \\ &= \mathbf{G}^\top \mathbf{a} = 1, \end{aligned} \quad (26)$$

where $\mathbf{G}$ is the vector of dimension N .

5.2 Additional Constraint for Cumulative Distribution.

Since $\int_{x_1}^{x_N} (x - 1)f''(x)dx = \int_{x_1}^1 (x - 1)f''(x)dx + \int_1^{x_N} (x - 1)f''(x)dx = 0$ and $f'' > 0$, the constraint (25) has the role that the implied distribution f'' is roughly symmetrically shaped around 1. From the view point of this, we can add more artificial condition of monotonicity to f'' such that f'' is monotone increasing before 1 and monotone decreasing after 1, which does not hurt the accuracy of the recovering power of f'' . This condition dramat-

ically reduces the whipsaw and peak of f'' at the knots.⁵

We describe this in the sense of cumulative distributions. Consider the first order derivatives of f at x_i ,

$$f'(x_i) - f'(0) = \int_0^{x_i} f''(x)dx = \text{cdf}(x_i), i = 1, \dots, N, \quad (27)$$

since $f'(0) = -1$ for this relation, the cumulative distribution function satisfies $\text{cdf}(x) = f'(x) + 1$. The inflection point of $\text{cdf}(x)$ is the point of sign changes for the second derivatives of $f'(x) + 1$, which equals $f'''(x)$.

$$\begin{aligned} \frac{\partial^3 f}{\partial x^3} \Big|_{x_i} &\geq 0, \quad i = 1, \dots, l, \\ \frac{\partial^3 f}{\partial x^3} \Big|_{x_i} &< 0, \quad i = l + 1, \dots, N, \end{aligned} \quad (28)$$

where $|x_l - 1| < |x_i - 1|$, $l \neq i$, $i = 1, \dots, N$.

5.3 Quadratic Programming for Implied Distribution.

Adding the three new conditions (23), (25) and (28) as well as removing the smoothness term in (22) ($\lambda = 0$) creates our new optimization problem. We sequentially calculate the call option price curves from time 0 (which is the payoff) to time t_M . However at this time, one must note that each of the call option price curves is used as a volatility surface that eventually aims to calculate the local volatility surface. From the view point of the discretized process (2), which starts from $t_0 = 0$ with a unique volatility σ_0 , the call option prices, which have different implied volatilities at time t_1 need local volatilities at the mid point of the time interval $[0, t_1]$, let this be denoted by $t_{1/2}$ and the volatility, which transfers the Arrow-Debreu price (dirac delta function) at the origin to each point of the forward moneyness at time $t_{1/2}$, is σ_0 .⁶ Now let's define new 2-dimensional mesh structure $\mathcal{T}$ which is not equispaced but is necessary for the calibration of call option price curves.

⁵This condition, which gives smoothing effects on the distribution, is very important to feasible calculation of the local volatilities from the call option prices or the implied distribution f'' itself. More specifically, in [9] uses implied distribution as the building block of their finite difference method and in [10] use call option prices as their finite difference.

⁶ σ_0 must satisfy $\text{call}(x, t_{1/2} | \sigma_0) < \text{call}(x, t_1)$ which does not violate the calendar arbitrage. Instead, when calculating the local volatilities, σ_0 must be high enough for possible reconstruction of call option prices at t_1 starting with local volatilities at $t_{1/2}$. This is important for pricing with local volatility, especially when using the monte-carlo method.

Table 3: Results of implied distributions with[rec.call1] and without[rec.call2] monotone-constraints (28) under QP (29): S&P500, 07/11/2019. QP with monotone constraint is slightly inaccurate but ignorable in most cases. We ignore the option price lower than 0.0001.

Time	F.mon	mk.call	rec.call1	rec.call2	mk.vol	rec.vol1	rec.vol2
1.112	0.5496	0.4535	0.4535	0.4535	0.3090	0.3090	0.3090
1.112	0.8994	0.1381	0.1381	0.1381	0.1948	0.1948	0.1948
1.112	1.0993	0.0227	0.0229	0.0227	0.1354	0.1359	0.1354

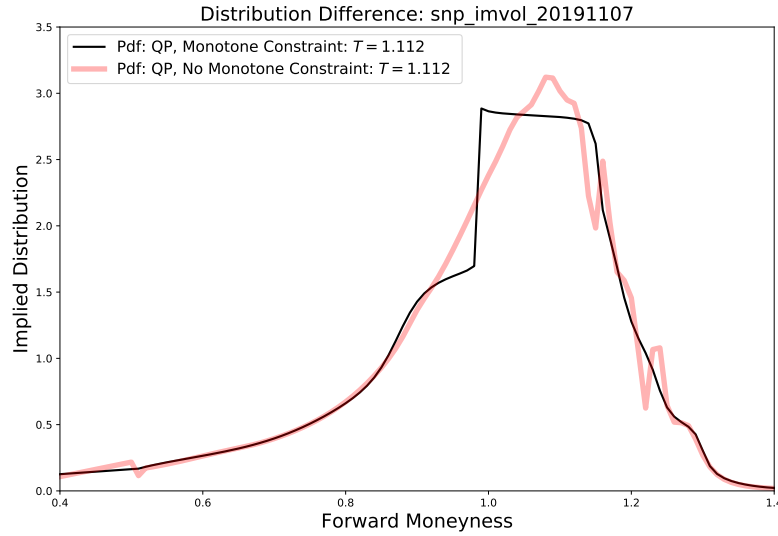


Figure 6: Two Implied Distributions with and without(blurred red line) monotone-constraints (28) under QP (29): S&P500, 07/11/2019

Let x_{min}, x_{max} be the pre defined min and max values of the forward moneyness and $x_{1,j}, \dots, x_{n,j}$ be the forward moneyness values corresponds to the regular exercise prices at maturity t_j ⁷. Subdivide the two line segments $[x_{min}, x_{1,j}]$ and $[x_{n,j}, x_{max}]$ with number $m > 0$ such that $\frac{x_{min}}{m}$ and $\frac{x_{max}}{m}$ are larger than each of $\Delta x_{i,j} = x_{i+1,j} - x_{i,j}, i = 1, \dots, n - 1$. Since the second part of the subscription in $x_{i,j}$ implies time t_j , we omit time for the next descriptions. Let this subdivision be $\mathcal{T}_j = [x_{1,j}, \dots, x_{N,j}], N = n + 2m$ and $\mathcal{T} = \mathcal{T}_{1/2} \cup \mathcal{T}_1 \cup \dots \cup \mathcal{T}_M$ is our mesh structure with $N \times (M + 1)$ size of location data. Figure 7 displays $\mathcal{T}$ -mesh structure. The variance $u_{i,j} = f_{TPS}(x_{i,j} | \Omega)$ is the result of the algorithm 1. Call option price

$y_{i,j} = \text{call}(x_{i,j}, u_{i,j}) = \text{call}(x_{i,j}, t_j, u_{i,j})$ in (10). Now the new quadratic optimization problem is as follows.

$$\begin{aligned}
& \min_{f_j} \left(\sum_{i=1}^N [w_{i,j}(y_{i,j} - f_j(x_{i,j}))]^2 \right), \text{ for } j = \frac{1}{2}, 1, \dots, M, \\
& \text{s.t. } f_j(x_{N,j}) \geq 0, \\
& \quad \frac{\partial f_j}{\partial x} \Big|_{x_{i,j}} \leq 0, \rightarrow (13a) \\
& \quad \frac{\partial^2 f_j}{\partial x^2} \Big|_{x_{i,j}} \geq 0, \quad i = 1, \dots, N, \rightarrow (13b) \\
& \quad f_{j-\alpha}(x_{i,j}) < f_j(x_{i,j}), \alpha = \frac{1}{2} \text{ when } j < 2, \alpha = 1 \text{ o.w.}, \rightarrow (13c) \\
& \quad \int_{x_{1,j}}^{x_{N,j}} f_j''(x) dx = 1, \rightarrow (23) \\
& \quad \int_{x_{1,j}}^{x_{N,j}} x f_j''(x) dx = 1, \rightarrow (25) \\
& \quad \frac{\partial^3 f_j}{\partial x^3} \Big|_{x_{i,j}} \geq 0, \quad i = 1, \dots, l, \quad \frac{\partial^3 f_j}{\partial x^3} \Big|_{x_{i,j}} \leq 0, \quad i = l+1, \dots, N \rightarrow (28)
\end{aligned} \tag{29}$$

where the time 0 call option price is $f_0(x) = \max(1-x, 0)$ and the index $l = \underset{i}{\operatorname{argmin}} |x_{i,j} - 1|$. Since $\frac{\partial^3 f_j}{\partial x^3} = 6d_i$ at $x = x_{i,j}, i = 1, \dots, N'$ for each t_j , the last two condition can be expressed as $\mathbf{D}_1 \mathbf{a} \geq \mathbf{0}$ and $\mathbf{D}_2 \mathbf{a} < \mathbf{0}$ where $\mathbf{D}_1$ is a sub matrix of $\mathbf{D}$ consisting of the 1 to l -th row and $\mathbf{D}_2$ is made up of the remaining rows.

5.4 Accurate Price Recovering from Implied Distribution.

After the optimization, we obtain a series of cubic spline coefficients which are used to recover the call option price curves as well as implied distributions. We can easily confirm the quality of the implied distributions by calculating the call option prices from the distribution using the simple integration rule for polynomials of order 2. For ease of notation we omit the

⁷In our experiment, allotting the same moneyness value $x_{i,0}$ to $x_{i,j}, j = 0, \dots, M$, for each $i = 1, \dots, n$, gives us accurate enough results with $dx = 0.01$ in $[0.5, 1.5]$ and 0.05 in outer region with $x_{min} = 0.0, x_{max} = e^{\beta \bar{\sigma} \sqrt{t_M}}$, where $\bar{\sigma}$ is market average volatility with $\beta - \sigma$ rule. Since our TPS approximate the range of moneyness about $0.4 \sim 1.6$, the outside values are fixed with these endpoints. In most cases corresponding $y_{max,j}$ of $x_{max} = 5$ is enough when $t_j \leq 5$.

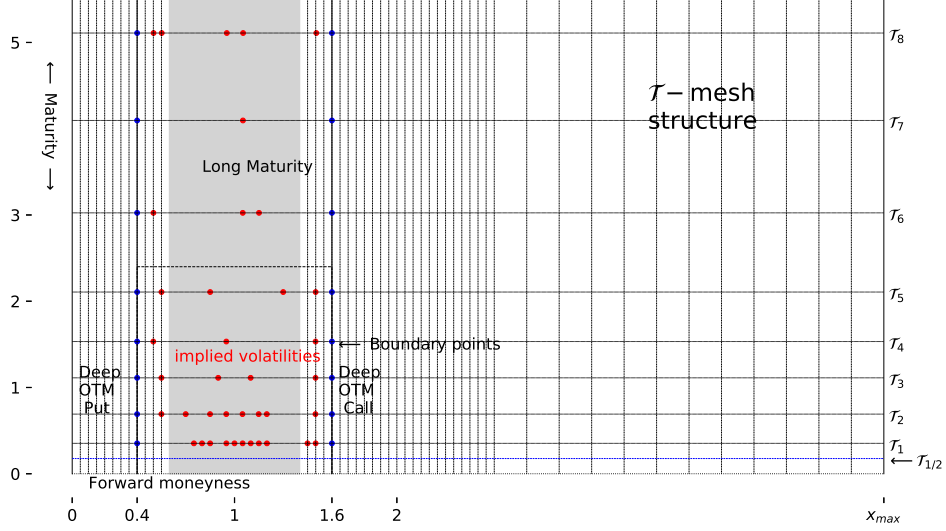


Figure 7: $\mathcal{T}$ -mesh structure: $x = [0, 5]$ is divided into 128 number of intervals with three different sizes, Δx_i 's of interval $[0.6, 1.4]$ are 0.01, others are 0.05 and 0.2.

j 's in $x_{i,j} \in \mathcal{T}_j$, for call option with exercise x_k in $\mathcal{S}$.

$$\begin{aligned}
 call(x_k, t_j) &= \int_{x_1}^{x_N} \max(x - x_k, 0) f''(x) dx, \\
 &= \sum_{i=k}^{N-1} \int_{x_i}^{x_{i+1}} (x - x_k) l(x | 2c_i, 2c_{i+1}) dx \longrightarrow P_i(x) := (x - x_k) l(x | 2c_i, 2c_{i+1}), \\
 &= \sum_{i=k}^{N-1} \frac{x_{i+1} - x_i}{6} \left[P_i(x_i) + 4P_i\left(\frac{x_{i+1} + x_i}{2}\right) + P_i(x_{i+1}) \right] \longrightarrow \text{simpson's rule}, \\
 &= \sum_{i=\max(k,2)}^{N-1} \frac{1}{3} [-x_{i-1}^2 + x_{i+1}^2 - x_{i-1}x_i + x_i x_{i+1} + x_{i-1}x_k - x_{i+1}x_k] c_i,
 \end{aligned} \tag{30}$$

where $k = 1, \dots, N-1$ and $l(x | 2c_i, 2c_{i+1})$ is a linear segment which connecting $(x_i, 2c_i)$ to $(x_{i+1}, 2c_{i+1})$. In matrix form, $\frac{1}{3} \mathbf{C} \mathbf{c} = \mathbf{C} \mathbf{S} \mathbf{a} = \mathbf{call}$, where $\mathbf{S}$ in (18), $\mathbf{call}$ is the call option price vector with k -th element $call(x_k, t_j)$ and $\mathbf{C}$ is the $(N-1) \times N$ matrix whose (k, i) element is the underlined term in (30).

5.5 Linear Programming for Implied Distribution.

We can construct a more direct optimization problem for the distribution f'' based on equation (30). In which, the objective function part is ,

$$\min_{\mathbf{a}} \sum_{i=1}^N |w_i(y_i - (\mathbf{C}\mathbf{S}\mathbf{a})_i)|,$$

where $(\mathbf{C}\mathbf{S}\mathbf{a})_i$ indicates the i -th element of the vector $\mathbf{C}\mathbf{S}\mathbf{a}$. Introducing the new slack variables $z_i = |w_i(y_i - (\mathbf{C}\mathbf{S}\mathbf{a})_i)|, i = 1, \dots, N-1$, we can transform this into linear programming problem (31) and (35), w.l.o.g. let $w_i = 1$.

$$\begin{aligned} \min_{\mathbf{a}} \left(\sum_{i=1}^{N-1} z_i = [\mathbf{0}, \mathbf{1}] \cdot \begin{bmatrix} \mathbf{a} \\ \mathbf{z} \end{bmatrix} \right), \\ \text{s.t.} \quad -z_i \leq 0, \\ \quad -(\mathbf{C}\mathbf{S}\mathbf{a})_i - z_i \leq -y_i, \\ \quad +(\mathbf{C}\mathbf{S}\mathbf{a})_i - z_i \leq +y_i, \quad i = 1, \dots, N-1, \\ \text{the same constraints as in (29) apply,} \end{aligned} \tag{31}$$

This is the final destination in finding the implied distributions we have been exploring. This highly mechanical processes, which has now progressed to the present stage, gives extremely accurate reconstruction results and high quality transition density functions that are very useful both in practice and in various areas of research. This LP method has the distinction that the value of the implied distribution function is characterized according to each section's moneyness $\mathbf{x}$ because the recovering of the call option price is made in a section by section manner, these sections are divided by the knot points in $\mathbf{x}$, which are displayed in Figure 8. When comparing the two methods, using the QP (29) method tends to give a more smoothly connected shape 8. Instead, the LP is looks more like a piecewise linear function, but it is straightforward to check the level of distribution while being more accurate than QP. The reconstructed price does not show a significant difference compared to the LP method 4. In algorithm 2, 3 the QP method is named CSQP and LP is CSLP. We test LP's results with matlab's `linprog` built-in function(dual-simplex) and with `cvxopt.solvers.lp` in GLPK's (GNU Linear Programming Kit) primal or dual revised simplex method. All the results with the robustness of the case study shows GLPK is better than that of `linprog`.

5.6 Transformation to Spot Moneyiness.

Because of the advantages of notation and explanation, all processes are explained based on forward moneyiness. However, for practical use, conversion to spot moneyiness is necessary. This is possible without difficulty.

First, since $\frac{\partial^2 C}{\partial x^2} = df(T)\mathbf{P}(x, T)$, where x is spot moneyiness. Equation (23) converts to the following equation.

$$\int_{x_0}^{x_N} f''(x)dx = df(T) \quad (32)$$

Second, the risk neutral condition (25) converts to the following equation.

$$\mathbb{E}[X] = \int_{x_0}^{x_N} x f''(x)dx = forward(T), \quad (33)$$

Lastly the calendar arbitrage condition (13c) in optimization problem (29) converts to the following equation which corresponds to (5c).

$$f_{j-\alpha}(x_i/fwd_{j-\alpha,j}) < f_j(x_i), \alpha = \frac{1}{2} \text{ when } j < 2, \alpha = 1 \text{ o.w.}, \quad (34)$$

The quadratic optimization problem (29) is modified as following.

$$\begin{aligned} \min_{f_j} & \left(\sum_{i=1}^N [w_{i,j}(y_{i,j} - f_j(x_i))]^2 \right), \text{ for } j = \frac{1}{2}, 1, \dots, M, \\ \text{s.t. } & f_j(0+) = qf_j, \\ & \int_{x_0}^{x_N} f_j''(x)dx = df_j, \quad \int_{x_0}^{x_N} x f_j''(x)dx = fwd_j, \\ & f_{j-\alpha}(x_i/fwd_{j-\alpha,j}) < f_j(x_i), \alpha = \frac{1}{2} \text{ when } j < 2, \alpha = 1 \text{ o.w.}, \\ & \text{the other conditions are the same.} \end{aligned}$$

where $y_{i,j}$ is call option price and last condition $fwd_{j-\alpha,j} = qf(t-\tau, t)/df(t-\tau, t)$ is the forwarding factor where $t - \tau = T_{j-\alpha}$, $t = T_j$.

In figure 14, shows the CSLP calibrated results for forward money and spot moneyiness based reconstruction results for the call option time values.

Algorithm 2: Function: CSQP, CSLP

Data: $\mathbf{x}$ (location), $\mathbf{y}$ (call option price, using TPS approximated variance surface), $\mathbf{lb}$ (lower bounds),

$\mathbf{x} = (x_1, \dots, x_N)$, $\mathbf{y} = (y_1, \dots, y_N)$, $\mathbf{lb} = (lb_1, \dots, lb_N)$

Result: $\mathbf{cf}$ (coefficients of cubic spline interpolation)

Function CSQP($\mathbf{x}, \mathbf{y}, \mathbf{lb}, \mathbf{w}$):

calculate $\mathbf{S}, \mathbf{B}, \mathbf{F}, \mathbf{D} = \begin{bmatrix} \mathbf{D}_1 \\ \mathbf{D}_2 \end{bmatrix}$ as a function of $\Delta \mathbf{x}$

proc *quadratic programming*:

Find: $\mathbf{a} = \operatorname{argmin} \left(\frac{1}{2} \mathbf{a}^\top \mathbf{W} \mathbf{a} - \mathbf{y}^\top \mathbf{W} \mathbf{a} \right)$, $\mathbf{W} = \operatorname{diag}(\mathbf{w})$.

Subject to:

$\mathbf{I} \mathbf{a} > \mathbf{lb} \dots$ call price greater than $\mathbf{lb}$,

$\mathbf{B} \mathbf{a} < \mathbf{0} \dots$ negative $\frac{\partial c}{\partial x}$,

$\mathbf{S} \mathbf{a} > \mathbf{0} \dots$ positive $\frac{\partial^2 c}{\partial x^2}$,

$\mathbf{D}_1 \mathbf{a} < \mathbf{0} \dots$ left side of pdf $\rightarrow$ increasing,

$\mathbf{D}_2 \mathbf{a} > \mathbf{0} \dots$ right side of pdf $\rightarrow$ decreasing,

$6 \mathbf{F}^\top \mathbf{S} \mathbf{a} = 1 \dots$ integration of pdf = 1,

$\mathbf{G}^\top \mathbf{a} = 1 \dots$ forward price = 1,

$\mathbf{b} = \mathbf{B} \mathbf{a}$, $\mathbf{c} = 3 \mathbf{S} \mathbf{a}$, $\mathbf{d} = \mathbf{D} \mathbf{a}$

$\mathbf{cf} = [\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}]$;

Function CSLP($\mathbf{x}, \mathbf{y}, \mathbf{lb}, \mathbf{w} \leftarrow 1$):

proc *linear programming*:

Find: $\mathbf{a} = \operatorname{argmin} (z_1 + \dots + z_{N-1})$.

Subject to:

$$\left[\begin{array}{c|c} \mathbf{0} & -\mathbf{I} \\ \hline -\mathbf{C} \mathbf{S} & -\mathbf{I} \\ \hline \mathbf{C} \mathbf{S} & -\mathbf{I} \end{array} \right] \begin{bmatrix} \mathbf{a} \\ \mathbf{z} \end{bmatrix} \leq \begin{bmatrix} \mathbf{0} \\ -\mathbf{y} \\ \mathbf{y} \end{bmatrix} \quad (35)$$

same constraints as in (29)

$\mathbf{cf} = [\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}]$;

Algorithm 3: Procedure: Link CSLP (CSQP)

Data: $\sigma_0, \Omega, f(\cdot | \Omega), \mathbf{w}$.
Result: $\text{cubicspline}(x | \mathbf{cf}_j), j = 1/2, 1, \dots, M$
 Construct $\mathcal{T} = \mathcal{T}_{1/2} \cup \mathcal{T}_1 \cup \dots \cup \mathcal{T}_M$ with $\dim. N \times (M + 1)$
for $j = [1/2, 1, 2, \dots, M]$ **do**
 if $j \geq 1$ **then**
 $u_{i,j} \leftarrow f_{TPS}(x_{i,j} | \Omega)$
 $lb_i \leftarrow \text{cubicspline}(x_{i,j} | \mathbf{cf}_{j-\alpha}), \alpha = 1/2$ if $j = 1$ else 1,
 else
 $u_{i,j} \leftarrow \sigma_0^2 t_{1/2}$
 $lb_i \leftarrow \max(1 - x_{i,j}, 0)$
 end
 $y_{i,j} \leftarrow \text{call}(x_{i,j}, u_{i,j})$
 $\mathbf{x} = (x_{1,j}, \dots, x_{N,j})$
 $\mathbf{y} = (y_{1,j}, \dots, y_{N,j})$
 $\mathbf{lb} = (lb_1, \dots, lb_N)$
 $\mathbf{cf}_j = \text{CSLP}(\mathbf{x}, \mathbf{y}, \mathbf{lb}, \mathbf{w})$
end

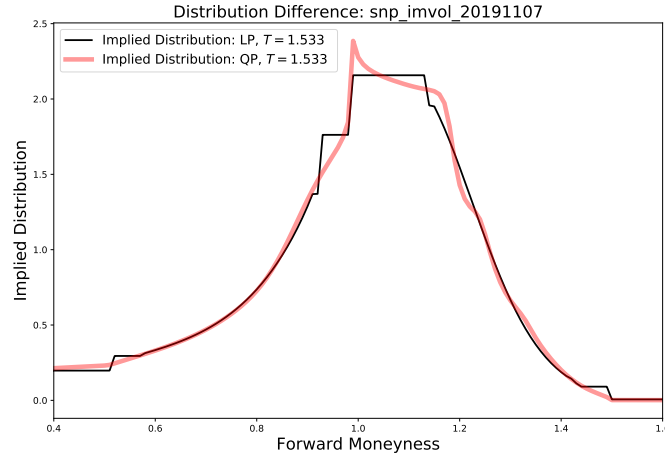


Figure 8: Implied Probability Distributions, with LP (31) and QP (29). S&P500
 Date: 07/11/2019, Using GLPK with Revised Simplex method.

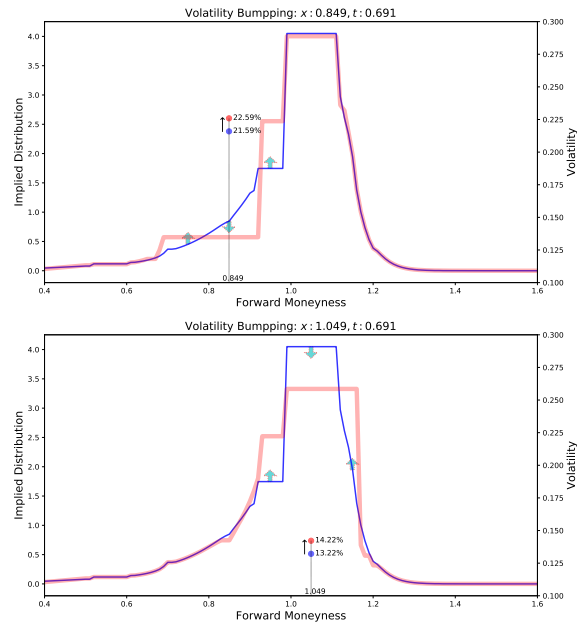


Figure 9: Volatility Bumping Effect. We can see the initial distribution spreading out both sides from the volatility shock which intensifying the diffusion. Original distribution: Blue colored solid line, Changed distribution: Blurred red line. Using GLPK with Revised Simplex method. S&P500, Date: 07/11/2019.

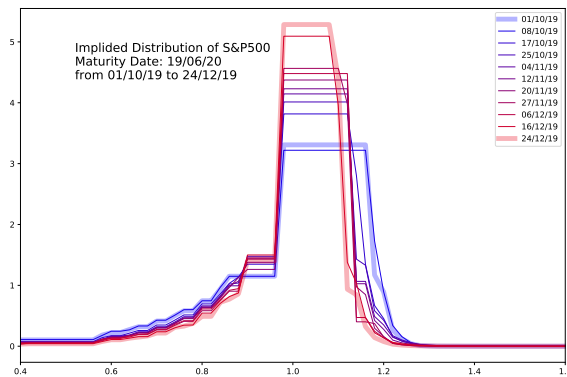


Figure 10: Time Series of implied distribution with Target Maturity(19/6/20), Dates from 01/10/19 to 24/12/19, S&P500

6 Conclusion.

The construction and application of volatility surfaces are essential for issuance and pricing of derivatives. This paper dealt with the problem of not only estimating the transition probability function inherent in the volatility information of the market but then also formulating an efficient approximation problem. The variance surface using the TPS is based on a simple assumption. “The black modes (1) in a market where the exercise price and maturity are close to each other are not very different.” The construction of the variance surface is an important task in the pre-smoothing stage, in processing of missing data, and in preventing calendar arbitrage. Next, we obtain a motif from the cubic spline call option price curve construction of Fengler [8], this is widely used as a call option price curve construction method that satisfies the NAC condition. This is used to explore the more fundamental problem of the construction condition of the transition probability function and optimization problem with implementations. As a problem of discretization instead as of the smoothing condition for the observed call option price, the monotone property of the distribution function was added to obtain a stable distribution function in which the distribution function eliminates the peak and whipsaw phenomenon at the knot. This can be considered as a kind of smoothing constraint. This condition may be artificial, but in terms of option pricing assuming risk neutrality, it may be a better option in many ways if there is no problem of price reconstruction. As our main result, we formulated a linear programming problem that approximates the call option prices using the distribution function itself. This method gives extremely accurate reliable results, which is confirmed by figure 8 (where it is compared using QP) and by the volatility bumping test. 9 The form of the analytically tractable transition distribution function clearly shows the concrete form of the risk neutral price distribution that is inherent in volatility information. In practice, it is expected that our method will play an important role in the calibration of stable local volatility as well as other financial engineering problems that need the implied transition distribution functions. We specified a sequential development process that starts with solving the missing data problem of the implied distributions, we also outlined related methods and presented the algorithm part separately promote its understanding. For the test, HSCEI (2015-11-9), which is an index with very little volatility data, and the S&P500 Index (2019-11-7), which collects part of the volatility data, are used to produce pictures and results tables that show the process and its results. The simple and clear method we propose is easily implemented in a program to check the proposed algorithm

and test various volatility data.

NO.	TIME	F.MON	MK.CALL	LP.CALL	QP.CALL
1	0.117	0.550	0.450	0.450	0.450
2	0.117	0.750	0.250	0.250	0.250
3	0.117	0.800	0.200	0.200	0.201
4	0.117	0.850	0.151	0.151	0.151
5	0.117	0.950	0.055	0.055	0.055
6	0.117	1.000	0.016	0.016	0.016
7	0.117	1.050	0.001	0.001	0.001
8	0.117	1.100	0.000	0.000	0.000
9	0.117	1.150	0.000	0.000	0.000
10	0.117	1.200	0.000	0.000	0.000
11	0.117	1.250	0.000	0.000	0.000
12	0.117	1.450	0.000	0.000	0.000
13	0.117	1.500	0.000	0.000	0.000
14	0.691	0.549	0.452	0.452	0.452
15	0.691	0.699	0.306	0.306	0.306
16	0.691	0.849	0.167	0.167	0.167
17	0.691	0.949	0.085	0.085	0.085
18	0.691	1.049	0.025	0.025	0.025
19	0.691	1.149	0.002	0.002	0.002
20	0.691	1.199	0.001	0.001	0.001
21	0.691	1.498	0.000	0.000	0.000
22	1.112	0.550	0.453	0.453	0.453
23	1.112	0.899	0.138	0.138	0.138
24	1.112	1.099	0.023	0.023	0.023
25	1.112	1.499	0.000	0.000	0.000
26	1.533	0.500	0.504	0.504	0.504
27	1.533	0.949	0.115	0.115	0.115
28	1.533	1.499	0.000	0.000	0.000
29	2.107	0.550	0.458	0.458	0.458
30	2.107	0.850	0.200	0.200	0.200
31	2.107	1.300	0.008	0.008	0.008
32	2.107	1.500	0.001	0.001	0.001
33	3.025	0.500	0.510	0.510	0.510
34	3.025	1.051	0.101	0.101	0.101
35	3.025	1.151	0.060	0.060	0.060
36	4.096	1.052	0.125	0.125	0.125
37	5.109	0.501	0.517	0.517	0.517
38	5.109	0.551	0.474	0.474	0.474
39	5.109	0.952	0.193	0.193	0.193
40	5.109	1.053	0.144	0.144	0.144
41	5.109	1.504	0.023	0.023	0.023

Table 4: Results of implied distributions for S&P500, 07/11/2019. There are no significant differences between the market and the two recovered values.

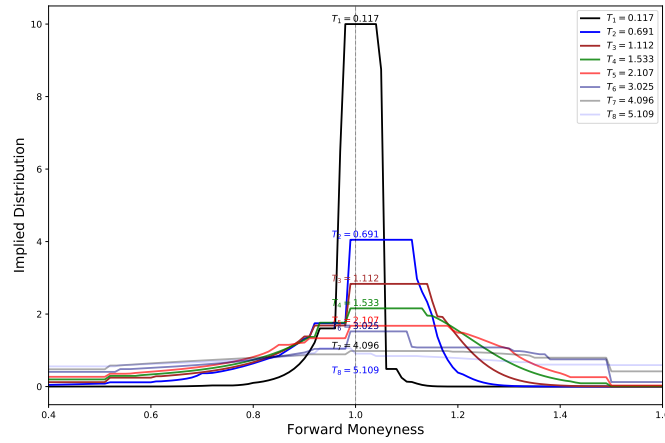


Figure 11: Series of Implied Distributions using LP (31), S&P500 Date: 07/11/2019

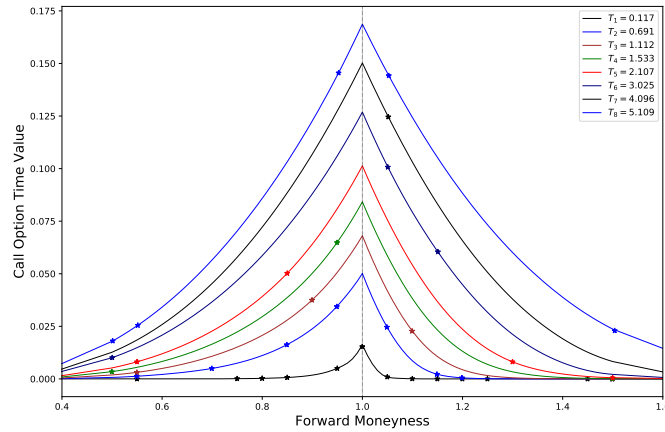


Figure 12: Recovered Call Option Time Values(-), $call(x, t) - \max(1 - x, 0)$ and Market Values(*) using LP (31) which are exactly the same. S&P500 Date: 07/11/2019

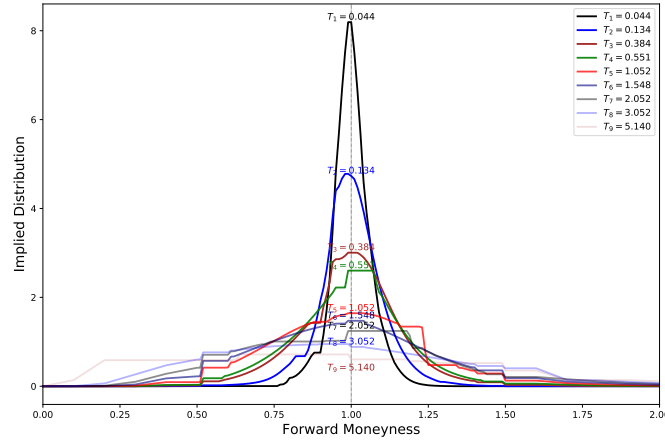


Figure 13: Implied Probability Distributions. HSCEI, Date: 9/11/2015 . Calibrated by CSLP method.

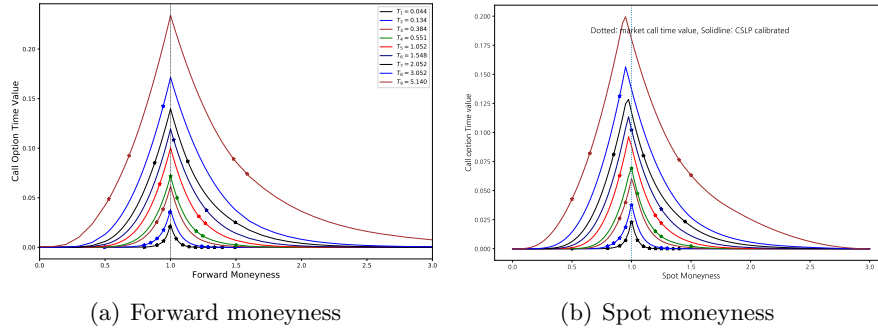


Figure 14: Recovered Call Option Times value results, Solid line: CSLP approximated and Dotted: Market Value, (*), (a) Forward moneyness, (b) Spot moneyness. 5.6 based calibration from data: HSCEI, Date: 9/11/2015

In figure (a), time value is given by $call(x, t, u) - \max(1 - x, 0)$ where x is forward moneyness and u is variance at (x, t) as for (10). In figure (b), time value is based on spot moneyness $x' = \frac{K}{S_0} = x \frac{q_f}{df}$ and is given by $call'(x', t, u') - \max(qf - x', 0)$,

$call'(x', t, u) = qfN(d_1) - dfx'N(d_2)$ where $d_1 = \frac{\ln(\frac{1}{x'})}{\sqrt{u'}} + \frac{1}{2}\sqrt{u'}$, $d_2 = d_1 - \sqrt{u'}$, $u' = \sigma^2 t$, σ is implied volatility at (x', t) , $df = e^{-\int_0^t r(s)ds}$, $dq = e^{-\int_0^t q(s)ds}$.

,

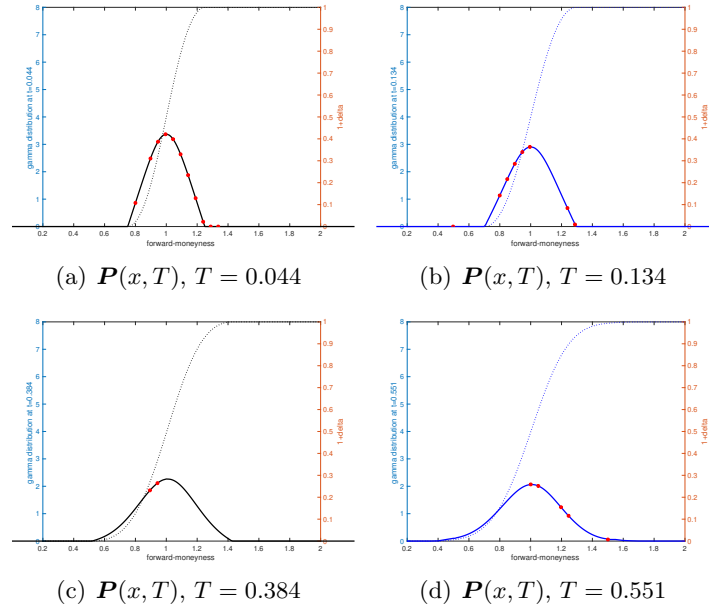


Figure 15: HSCEI Implied Distributions with Smoothing Parameter $p = 0.99$, Date: 9/11/2015, distribution is well shaped.

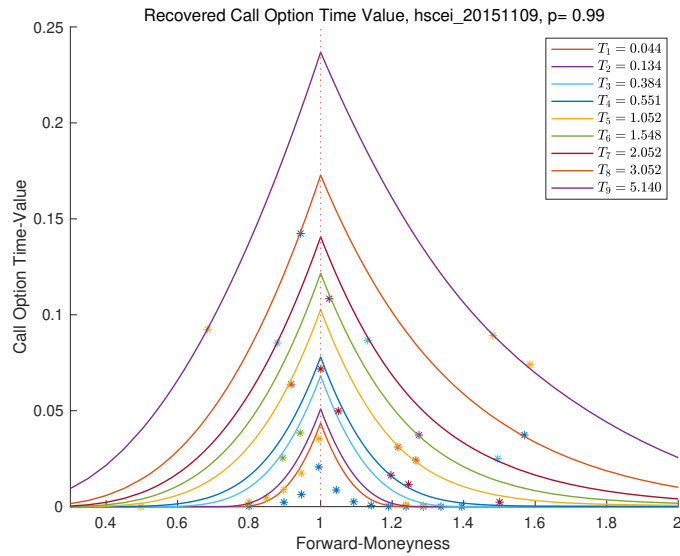


Figure 16: Call Option Time Values with Smoothing Parameter $p = 0.99$, $call(x, t) - \max(1 - x, 0)$, HSCEI, Date: 2015-11-9, Fail to recover for short maturities.

Appendix: RBF and TPS basis

A radial basis function(RBF) is a real-valued function $\varphi : \mathbf{R}^d \rightarrow \mathbf{R}$ whose value depends only on the distance from the origin. Therefore, for arbitrary point $\mathbf{x} \in \mathbf{R}^d$, $\varphi(\mathbf{x}) = \varphi(\|\mathbf{x}\|) = \varphi(\mathbf{r})$ holds. For $\mathbf{x} = (x, t)$ in dimension $d = 2$, $\mathbf{r} = \|\mathbf{x}\| = \sqrt{x^2 + t^2}$. Now we only consider the case $d = 2$ for simplifying the arguments. The most popular cubic spline interpolation needs gridded data(a mesh structure) for interpolation. However, arbitrary scattered data is not a gridded type of data. This is why the RBF is used for scattered data interpolation as it depends only on the distance from the origin. [16]

Let the observed data, which is the location and value pair, $(\mathbf{x}_i, y_i), \mathbf{x}_i \neq \mathbf{x}_j, i, j = 1, 2, \dots, N$, a function $f(\mathbf{x})$ denote the value of an unobserved point $\mathbf{x}$ of true but unknown function.

Let's consider the linear combination of RBFs which fits the data y_i at the given points

$$f_{RBF}(\mathbf{x}) = \sum_{i=1}^N c_i \varphi(\|\mathbf{x} - \mathbf{x}_i\|), i = 1, 2, \dots, N \quad (36)$$

The coefficient $c_i, i = 1, \dots, N$ can be obtained by directly substituting $\mathbf{x}$ to $\mathbf{x}_i$ in the above equation.

$$\mathbf{A} \cdot \mathbf{c} = \mathbf{y} \quad (37)$$

where $\mathbf{A} = (\varphi(\|\mathbf{x}_i - \mathbf{x}_j\|))_{i=1, \dots, N}^{j=1, \dots, N}, \mathbf{c} = (c_1, \dots, c_N)^\top, \mathbf{y} = (y_1, \dots, y_N)^\top$.

The solvability of this matrix equation is well known when the matrix $\mathbf{A}$ is positive definite. The radial basis function, which makes the symmetric matrix $\mathbf{A}$ solvable, can also be called a positive definite radial basis function.

.1 The extended form of RBF and its matrix equation.

The extended form of RBF approximation has been suggested by Fasshauer(ch 6.3) [16]. The original RBF approximation (36) can not approximate the data well as it forms a plane like flattened shape because of the nonlinear character of the basis function. In this case adding the linear basis, $(\mathbf{x}, 1) = (x, t, 1)$ to the original $\varphi(\|\mathbf{x} - \mathbf{x}_i\|), i = 1, 2, \dots, N$, is suggested in Fasshauer(ch 6.2) [16]

$$f_{RBF}(\mathbf{x}) = \sum_{j=1}^N c_j \varphi(\|\mathbf{x} - \mathbf{x}_j\|) + c_{N+1}x + c_{N+2}t + c_{N+3}1 \quad (38)$$

Theorem .1. Let $\mathbf{A}$ be a real symmetric $N \times N$ matrix that is conditionally positive definite of order one, and let $P = (X, T, \mathbf{1})$ be an $N \times 3$ data matrix, where $X = (x_1, \dots, x_N)^\top$, $T = (t_1, \dots, t_N)^\top$ and $\mathbf{1}$ is a $N \times 1$ vector where all elements are 1. Then the system of linear equations has a unique solution.

$$\begin{pmatrix} A & P \\ P^\top & \mathbf{O} \end{pmatrix} \cdot \begin{pmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{y} \\ \mathbf{0} \end{pmatrix} \quad (39)$$

where $A = (a_{ij})_{i,j=1,\dots,N}$, $a_{ij} = \varphi(\|\mathbf{x}_i - \mathbf{x}_j\|)$, $P = (\mathbf{X}_{1:N}, 1)$, $\mathbf{X}_{1:N} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$ is $(N \times 2)$ - data matrix.

Proof. Refer to Fasshauer [16]. □

The matrix $A_{i,j} = (\varphi(\|\mathbf{x}_i - \mathbf{x}_j\|))$ is called conditionally positive definite matrix under the condition of $c_{N+1}x + c_{N+2}t + c_{N+3} = 0$, where $\mathbf{x}_i = (x_i, t_i)$, $i = 1, \dots, N$.

This theorem is important, because even if the matrix $\mathbf{A}$ is not positive definite adding the additional basis $(x, t, 1)$ makes the equation (39) have a unique solution.

.2 Thin Plate Spline Basis.

In the problem of approximating the surface of scattered data, it is often required to approximate the surface with minimal curvature. The criterion for mathematically measuring the bending is to minimize the second derivatives of $f''(x)$ of the twice differentiable function $f(x)$ for any point x . In $\mathbf{x} = (x, t)$ coordinates, this value is expressed as follows over the entire area of $\mathbb{R}^2$.

$$\|f\|_{distort}^2 = \int_{\mathbb{R}^2} \|D^2 f(\mathbf{x})\|_{frob}^2 d\mathbf{x}, \quad (40)$$

where

$$\|D^2 f(\mathbf{x})\|_{frob}^2 = \left(\frac{\partial^2 f}{\partial x^2}\right)^2 + 2\left(\frac{\partial^2 f}{\partial x \partial t}\right)^2 + \left(\frac{\partial^2 f}{\partial t^2}\right)^2 \quad (41)$$

The poly-harmonic function $2\mathbf{r}^2 \ln(\mathbf{r})^8$ is called a Thin Plate Spline (Duchon [13]). The thin plate spline function is a fundamental solution of the Biharmonic equation (41).

⁸ $\mathbf{r} = \|\mathbf{x} - \mathbf{x}_i\|$, $\mathbf{x} = (x, t)$, $\mathbf{x}_i = (x_i, t_i)$, $i = 1, \dots, N$ are observed data

The thin plate spline approximation using the thin plate spline basis can not guarantee a unique solution for equation (37), because the matrix $\mathbf{A}$ is conditionally positive definite matrix, but by adding the basis $(x, t, 1)$, the extended matrix equation (39) has a unique solution. (Fasshauer p64 Theorem 7.2 [16])

2.1 TPS and Penalized Least Squares Fitting

The problem of minimizing the distortion of the surface while fitting N scattered points in 2D space considering the importance of each point is the least square form of $\|f\|_{\text{distort}}^2$ with the penalty term $(1-p)\|f\|_{\text{distort}}^2$.

$$J(f) = \min_f \left[p \sum_{i=1}^N w_i (y_i - f(\mathbf{x}_i))^2 + (1-p)\|f\|_{\text{distort}}^2 \right] \quad (42)$$

where $w_i, i = 1, \dots, N$ is the weight for the importance of each point $\mathbf{x}_i$, and p ($0 \leq p \leq 1$) is the trade-off between the penalty and fitness term. The norm $\|\cdot\|_{\text{distort}}^2$ which measures the bending in the region $\mathcal{R}$ is $\|f\|_{\text{distort}}^2 = \int_{\mathcal{R}} \|D^2 f(\mathbf{x})\|_{f_{rob}}^2 d\mathbf{x}$, where $D^2 f$ is the Jacobian matrix of f w.r.t each variables i.e $\|D^2 f(\mathbf{x})\|_{f_{rob}}^2$ is the square of the sum of the all second derivatives of f .

Let f_{TPS} be a TPS approximation function with basis $\{\varphi_1, \dots, \varphi_n, x, t, 1\}$, $\varphi_i = \varphi(\|\mathbf{x} - \mathbf{x}_i\|) = 2\|\mathbf{x} - \mathbf{x}_i\|^2 \ln(\|\mathbf{x} - \mathbf{x}_i\|)$. The problem in finding f_{TPS} for all observation points is to find the coefficient $\mathbf{c} = \{c_1, \dots, c_{n+3}\}$ for the expression of $f_{TPS}(x, t) = \varphi_1 c_1 + \dots + \varphi_n c_n + x c_{n+1} + t c_{n+2} + c_{n+3}$ under the condition of $J(f_{TPS})$ (42).

Since $f_{TPS}(x_i, t_i) = \varphi_{i,1} c_1 + \dots + \varphi_{i,n} c_n + x_i c_{n+1} + t_i c_{n+2} + c_{n+3}$, and w.l.o.g set $w_i = 1$ in (42), let $n \times n$ matrix: $\Phi_1 = (\varphi_{i,j})_{1 \leq i,j \leq n}$, $\varphi_{i,j} = \varphi(\|\mathbf{x}_i - \mathbf{x}_j\|)$, $n \times (n+3)$ matrix: $\Phi = [\Phi_1, \mathbf{P}]$, $n \times 3$ matrix: $\mathbf{P} = [\mathbf{x}^*, \mathbf{t}^*, \mathbf{1}]$, $\mathbf{x}^* = (x_1, \dots, x_n)^\top$, $\mathbf{t}^* = (t_1, \dots, t_n)^\top$, $\mathbf{1} = \mathbf{1}_{n \times 1}$.

$$\sum_{i=1}^n (\varphi_{i,1} c_1 + \dots + \varphi_{i,n} c_n + x_i c_{n+1} + t_i c_{n+2} + c_{n+3} - y_i)^2 = \|\Phi \mathbf{c} - \mathbf{y}\|^2,$$

From Eberly [15],

$$\int_{\mathcal{R}} \left(\frac{\partial^2 f}{\partial x^2} \right)^2 + 2 \left(\frac{\partial^2 f}{\partial x \partial t} \right)^2 + \left(\frac{\partial^2 f}{\partial t^2} \right)^2 dx dt = (1-p) \mathbf{c}_1^\top \Phi_1 \mathbf{c}_1,$$

holds.

We now express $J(f_{TPS})$ by the matrix form,

$$\begin{aligned} J(f_{TPS}, p) &= \min_{\mathbf{c}} \left\{ p \|\Phi \mathbf{c} - \mathbf{y}\|^2 + (1-p) \mathbf{c}_1^\top \Phi_1 \mathbf{c}_1 \right\} \\ &= \min_{\mathbf{c}} \left\{ \mathbf{c}^\top (p\mathbf{A} + (1-p)\mathbf{B}) \mathbf{c} - 2p \mathbf{y}^\top \Phi \mathbf{c} \right\} + \mathbf{y}^\top \mathbf{y}, \end{aligned} \quad (43)$$

where $\mathbf{A} = \Phi^\top \Phi$, $\mathbf{B} = \begin{cases} \Phi_1, & 1 \leq i, j \leq n \\ 0, & n+1 \leq i, j \leq n+3 \end{cases}$ and $\mathbf{P}^\top \mathbf{c}_1 = 0$.

.3 Thin Plate Smoothing Spline.

If there are no other constraints, equation (43) can be solved using an FOC(First Order Condition). That is, differentiate it by $\mathbf{c}$ and construct the matrix equation.

$$\begin{pmatrix} \left(\Phi_1 + \frac{1-p}{p} I \right) & \mathbf{P} \\ \mathbf{P}^\top & 0 \end{pmatrix} \cdot \begin{pmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{y} \\ 0 \end{pmatrix} \quad (44)$$

Algorithm 4: General TPS Approximation Solver

Data: Data set $\Omega_N = \{(\mathbf{x}_i, y_i) \mid i = 1, \dots, N\}$, $\mathbf{x}_i = (x_i, t_i)$, y_i : i -th data location and value.

Proc: Make Kernel Matrix $\Phi_1 = (\varphi_{i,j})_{1 \leq i,j \leq n}$ and $\mathbf{P} = [\mathbf{x}^*, \mathbf{t}^*, \mathbf{1}]$

Result: $\mathbf{c}_1, \mathbf{c}_2$

[1]: **QR-Decompose of $\mathbf{P}$;**

$$\mathbf{P} = \mathbf{Q}\mathbf{R} = (\mathbf{Q}_1 : \mathbf{Q}_2) \begin{pmatrix} \mathbf{R}_1 \\ \mathbf{O} \end{pmatrix} \Rightarrow \mathbf{P}^\top = \mathbf{R}_1^\top \mathbf{Q}_1^\top;$$

$\mathbf{Q} = (\mathbf{Q}_1 : \mathbf{Q}_2)$ is orthogonal matrix where $\mathbf{Q}_1$ is $n \times 3$, $\mathbf{Q}_2$ is $n \times (n-3)$;

$\mathbf{R}_1$ is 3×3 upper triangular, $\mathbf{O}$ is $n \times 3$ zero matrix;

[2]: **Calculation of $\mathbf{c}_1$;**

$\mathbf{P}^\top \mathbf{c}_1 = 0 \Rightarrow \mathbf{R}_1^\top \mathbf{Q}_1^\top \mathbf{c}_1 = 0 \Rightarrow \mathbf{c}_1 = \mathbf{Q}_2 \gamma$ where γ is some $n-3$ vector;

Let $\mathbf{M} = \left(\Phi_1 + \frac{1-p}{p} \mathbf{I} \right)$ multiplying $\mathbf{Q}_2^\top$ to the first row of (44)

$$\begin{aligned} \mathbf{Q}_2^\top \mathbf{M} \mathbf{c}_1 &= \mathbf{Q}_2^\top \mathbf{y} \\ \mathbf{c}_1 &= \mathbf{Q}_2 \gamma \Rightarrow \mathbf{Q}_2 (\mathbf{Q}_2^\top \mathbf{M} \mathbf{Q}_2)^{-1} \mathbf{Q}_2^\top \mathbf{y} \end{aligned} \quad (45)$$

[3]: **Calculation of $\mathbf{c}_2$;**

Multiplying $\mathbf{Q}_1^\top$ to the first row of (44)

$$\begin{aligned} \mathbf{Q}_1^\top \mathbf{y} &= \mathbf{Q}_1^\top \left(\Phi_1 + \frac{1-p}{p} \mathbf{I} \right) \mathbf{c}_1 + \mathbf{Q}_1^\top \mathbf{P} \mathbf{c}_2 \\ \mathbf{Q}_1^\top \mathbf{y} &= \mathbf{Q}_1^\top \Phi_1 \mathbf{c}_1 + \frac{1-p}{p} \mathbf{I} \mathbf{Q}_1^\top \mathbf{Q}_2 \gamma + \mathbf{Q}_1^\top \mathbf{Q}_1 \mathbf{R}_1 \mathbf{c}_2 \\ \mathbf{R}_1 \mathbf{c}_2 &= \mathbf{Q}_1^\top (\mathbf{y} - \Phi_1 \mathbf{c}_1) \end{aligned} \quad (46)$$

$\mathbf{c}_2$ can be easily calculated using the LU decomposition, for $\mathbf{R}_1$ is upper triangular matrix;

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